Requirement Analysis Document

Project Name: Maze Solver Game

Course Name: Software Development 1

Programming Language: Java

Prepared By: Hridita Isalam

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1. Introduction

1.1 Project Overview

Maze Solver Game is a Java-based software application that allows users to navigate through a maze and reach a destination point. The software provides both manual gameplay and an automatic maze-solving feature. Users can explore the maze themselves or use the built-in solver to find the correct path.

1.2 Purpose

The purpose of this project is to develop an interactive software application that demonstrates problem-solving techniques, pathfinding algorithms, graphical user interface development, and object-oriented programming concepts.

1.3 Objectives

- Develop a user-friendly maze-solving application.
- Allow manual maze navigation.
- Implement automatic maze-solving functionality.
- Visualize the maze and solution path.
- Apply software development principles and OOP concepts.

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2. Functional Requirements

FR-1: Start Game

The system shall allow users to start a new maze-solving session.

FR-2: Display Maze

The system shall display a maze consisting of walls, paths, a starting point, and a destination point.

FR-3: Player Movement

The system shall allow users to move through the maze using keyboard controls.

FR-4: Movement Validation

The system shall prevent users from moving through walls or outside the maze boundaries.

FR-5: Move Counter

The system shall count and display the total number of moves made by the user.

FR-6: Game Timer

The system shall display the elapsed time during gameplay.

FR-7: Restart Game

The system shall provide an option to restart the maze at any time.

FR-8: Auto Solver

The system shall automatically find a valid path from the start point to the destination using a maze-solving algorithm.

FR-9: Path Visualization

The system shall visually display the solution path generated by the Auto Solver.

FR-10: Win Detection

The system shall detect when the player reaches the destination and display a success message.

FR-11: About Section

The system shall provide information about the software and its developer.

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3. Non-Functional Requirements

NFR-1: Usability

The software should provide a simple and easy-to-use interface.

NFR-2: Performance

The system should respond to user actions without noticeable delay.

NFR-3: Reliability

The application should operate without crashes during normal use.

NFR-4: Maintainability

The source code should be modular and organized using object-oriented programming principles.

NFR-5: Portability

The software should run on any system that supports Java Runtime Environment (JRE).

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4. System Requirements

Hardware Requirements

- Processor: Dual-Core Processor or Higher
- RAM: 4 GB or Higher
- Storage: 100 MB Free Space

Software Requirements

- Java Development Kit (JDK 17 or later)
- Java Swing Library
- Git
- GitHub

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5. Assumptions and Constraints

Assumptions

- Users have basic knowledge of keyboard controls.
- Java Runtime Environment is installed on the target system.

Constraints

- The project will be developed using Java.
- The project duration is limited to 10 weeks.
- Development will follow the Software Development course guidelines.

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6. Expected Outcome

The final software will provide an interactive maze-solving experience where users can manually navigate through a maze or use an automatic solver to find the correct path. The application will include a graphical user interface, move tracking, timer functionality, path visualization, and win detection.

Conclusion

Maze Solver Game is designed to demonstrate software development practices, object-oriented programming, and algorithmic problem solving. The project will provide both educational value and an engaging user experience while fulfilling the requirements of the Software Development course.